

## GIELAS, Sweden

WMO: 021100

Lat: 65.327N

Lon: 15.065E

Elev: 1893

StdP: 13.72

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-29.0 | (c)<br>-23.2 | (d)<br>-35.2                  | (e)<br>0.8 | (f)<br>-29.0 | (g)<br>-28.7 | (h)<br>1.2 | (i)<br>-23.2 | (j)<br>17.7           | (k)<br>32.4 | (l)<br>15.6 | (m)<br>26.2 | (n)<br>2.2            | (o)<br>250 | (p)<br>0.725 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>19.7            | (c)<br>72.9     | (d)<br>59.5 | (e)<br>69.3 | (f)<br>57.9 | (g)<br>65.4 | (h)<br>56.4 | (i)<br>61.9         | (j)<br>70.0 | (k)<br>59.6 | (l)<br>66.8 | (m)<br>57.4 | (n)<br>64.0 | (o)<br>5.4           | (p)<br>200 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.7                            | 79.1 | 65.3 | 56.4 | 72.6 | 62.4 | 54.3 | 67.1 | 60.5 | 28.6          | 70.7 | 27.0 | 66.7 | 25.5 | 64.0 | 68.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 14.4          | (b) 12.2 | (c) 10.6 | (e) -36.3                  | (f) 75.9 | (g) 4.3            | (h) 5.6  | (i) -39.4                                          | (j) 79.9  | (k) -41.9  | (l) 83.2  | (m) -44.3  | (n) 86.3  | (o) -47.4  | (p) 90.4  |    |    |
| (4) 14.4          | (5) 12.2 | (6) 10.6 | (7) -37.2                  | (8) 63.8 | (9) 4.1            | (10) 2.9 | (11) -40.1                                         | (12) 65.9 | (13) -42.6 | (14) 67.6 | (15) -44.9 | (16) 69.3 | (17) -47.9 | (18) 71.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.1   | 11.1  | 11.8  | 18.2  | 29.0  | 38.6  | 47.6  | 53.2  | 49.8  | 42.6  | 32.3 | 21.1  | 16.6  |  |  |
|                                                                        | DBStd       | 17.72  | 15.21 | 14.73 | 11.92 | 7.89  | 6.10  | 6.01  | 5.64  | 5.25  | 5.72  | 7.59 | 12.24 | 14.19 |  |  |
|                                                                        | HDD50       | 7142   | 1205  | 1070  | 987   | 629   | 358   | 117   | 31    | 68    | 228   | 548  | 866   | 1035  |  |  |
|                                                                        | HDD65       | 12372  | 1670  | 1490  | 1452  | 1079  | 818   | 522   | 368   | 471   | 673   | 1013 | 1316  | 1500  |  |  |
|                                                                        | CDD50       | 247    | 0     | 0     | 0     | 0     | 5     | 45    | 130   | 62    | 5     | 0    | 0     | 0     |  |  |
|                                                                        | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0    | 0     | 0     |  |  |
|                                                                        | CDH74       | 68     | 0     | 0     | 0     | 0     | 0     | 15    | 50    | 3     | 0     | 0    | 0     | 0     |  |  |
|                                                                        | CDH80       | 5      | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 0     | 0     | 0    | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.6    | 4.4   | 4.4   | 5.1   | 4.9   | 5.0   | 4.8   | 4.1   | 3.8   | 4.5   | 4.7  | 4.1   | 4.6   |  |  |
|                                                                        | PrecAvg     | 27.3   | 2.5   | 2.2   | 1.9   | 1.3   | 1.5   | 2.5   | 3.7   | 3.0   | 2.2   | 2.1  | 2.0   | 2.4   |  |  |
| (15) Precipitation                                                     | PrecMax     | 37.2   | 4.8   | 6.5   | 4.5   | 2.3   | 3.2   | 5.0   | 6.9   | 6.6   | 5.9   | 3.4  | 4.4   | 4.1   |  |  |
|                                                                        | PrecMin     | 21.3   | 0.4   | 0.3   | 0.3   | 0.2   | 0.5   | 0.4   | 1.4   | 0.6   | 0.7   | 0.9  | 0.6   | 0.9   |  |  |
|                                                                        | PrecStd     | 5.1    | 1.1   | 1.5   | 1.0   | 0.6   | 0.7   | 1.1   | 1.6   | 1.7   | 1.1   | 0.7  | 1.0   | 1.1   |  |  |
|                                                                        | PrecStd     | 5.1    | 1.1   | 1.5   | 1.0   | 0.6   | 0.7   | 1.1   | 1.6   | 1.7   | 1.1   | 0.7  | 1.0   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 38.9   | 39.8  | 42.5  | 54.6  | 69.1  | 76.3  | 79.4  | 74.1  | 64.1  | 51.7  | 45.5 | 39.9  |       |  |  |
|                                                                        | 0.4% MCWB   | 36.8   | 37.0  | 37.4  | 43.0  | 53.2  | 59.6  | 62.7  | 61.0  | 52.7  | 47.3  | 44.1 | 37.7  |       |  |  |
|                                                                        | 2% DB       | 35.5   | 36.8  | 38.7  | 47.9  | 62.3  | 70.7  | 74.1  | 69.9  | 58.6  | 47.8  | 40.8 | 37.6  |       |  |  |
|                                                                        | 2% MCWB     | 33.9   | 34.9  | 35.2  | 39.7  | 50.2  | 55.6  | 60.2  | 59.7  | 51.7  | 44.2  | 38.8 | 36.0  |       |  |  |
|                                                                        | 5% DB       | 33.2   | 34.1  | 36.7  | 43.7  | 56.6  | 65.7  | 70.4  | 65.8  | 55.1  | 45.2  | 37.5 | 35.6  |       |  |  |
|                                                                        | 5% MCWB     | 31.9   | 32.5  | 33.9  | 38.1  | 47.7  | 54.0  | 58.9  | 58.0  | 50.0  | 42.4  | 35.6 | 34.1  |       |  |  |
|                                                                        | 10% DB      | 29.8   | 31.1  | 34.6  | 40.6  | 50.5  | 61.1  | 66.5  | 61.6  | 52.5  | 43.0  | 34.9 | 33.3  |       |  |  |
|                                                                        | 10% MCWB    | 28.6   | 29.6  | 32.1  | 36.3  | 44.1  | 51.9  | 57.9  | 55.4  | 48.7  | 40.8  | 33.5 | 31.8  |       |  |  |
|                                                                        | 0.4% WB     | 36.9   | 37.3  | 38.0  | 44.1  | 54.5  | 62.4  | 65.1  | 64.3  | 55.6  | 48.4  | 43.5 | 37.9  |       |  |  |
|                                                                        | 0.4% MCDB   | 38.8   | 39.4  | 40.9  | 52.0  | 66.4  | 72.6  | 73.5  | 70.2  | 60.9  | 51.2  | 44.6 | 39.3  |       |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 34.1   | 35.2  | 35.7  | 41.2  | 51.3  | 58.3  | 62.7  | 61.0  | 52.8  | 45.0  | 38.4 | 36.2  |       |  |  |
|                                                                        | 2% MCDB     | 35.3   | 36.8  | 38.1  | 46.4  | 59.8  | 67.2  | 72.2  | 67.9  | 56.9  | 47.2  | 40.1 | 37.5  |       |  |  |
|                                                                        | 5% WB       | 32.0   | 32.8  | 34.1  | 38.6  | 48.1  | 55.3  | 60.5  | 58.7  | 50.8  | 43.1  | 35.5 | 34.1  |       |  |  |
|                                                                        | 5% MCDB     | 33.3   | 34.1  | 36.5  | 42.7  | 56.6  | 63.5  | 67.6  | 64.8  | 54.4  | 45.2  | 36.8 | 35.4  |       |  |  |
|                                                                        | 10% WB      | 28.8   | 29.5  | 32.2  | 36.5  | 44.6  | 52.7  | 58.5  | 56.3  | 49.1  | 41.1  | 33.5 | 31.9  |       |  |  |
|                                                                        | 10% MCDB    | 29.8   | 30.9  | 34.6  | 40.4  | 50.8  | 59.5  | 65.4  | 61.5  | 52.1  | 43.1  | 34.7 | 33.0  |       |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 16.6   | 18.7  | 21.0  | 17.8  | 16.8  | 18.9  | 19.7  | 18.9  | 15.5  | 11.3  | 12.6 | 15.1  |       |  |  |
|                                                                        | 5% DB       | 12.3   | 12.2  | 16.8  | 22.5  | 29.3  | 29.4  | 28.5  | 27.6  | 21.0  | 13.4  | 10.3 | 12.9  |       |  |  |
|                                                                        | 5% WB       | 11.7   | 10.7  | 14.0  | 15.8  | 18.9  | 16.9  | 15.8  | 18.3  | 14.2  | 10.5  | 9.4  | 12.0  |       |  |  |
|                                                                        | MCDBR       | 11.6   | 12.4  | 14.6  | 18.5  | 27.7  | 25.1  | 25.0  | 24.2  | 18.8  | 12.7  | 10.2 | 13.1  |       |  |  |
|                                                                        | MCWBR       | 11.1   | 11.2  | 12.4  | 13.5  | 18.9  | 16.7  | 16.1  | 17.0  | 13.2  | 10.1  | 9.4  | 12.1  |       |  |  |
|                                                                        | MCWBR       | 11.1   | 11.2  | 12.4  | 13.5  | 18.9  | 16.7  | 16.1  | 17.0  | 13.2  | 10.1  | 9.4  | 12.1  |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.174  | 0.246 | 0.278 | 0.306 | 0.316 | 0.317 | 0.328 | 0.315 | 0.291 | 0.269 | N/A  | 0.177 |       |  |  |
|                                                                        | taud        | 1.929  | 2.181 | 2.230 | 2.260 | 2.417 | 2.518 | 2.504 | 2.526 | 2.578 | 2.412 | N/A  | 1.941 |       |  |  |
|                                                                        | Ebn at Noon | 108    | 208   | 254   | 274   | 283   | 285   | 278   | 272   | 256   | 203   | N/A  | 26    |       |  |  |
|                                                                        | Edh at Noon | 7      | 17    | 26    | 32    | 31    | 29    | 28    | 25    | 19    | 14    | N/A  | 2     |       |  |  |
|                                                                        | Edh at Noon | 7      | 17    | 26    | 32    | 31    | 29    | 28    | 25    | 19    | 14    | N/A  | 2     |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 32     | 174   | 612   | 1195  | 1501  | 1544  | 1455  | 1088  | 594   | 222   | 56   | 9     |       |  |  |
|                                                                        | RadStd      | 3      | 37    | 73    | 97    | 112   | 141   | 179   | 115   | 75    | 26    | 7    | 0     |       |  |  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (37 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page